Neural Networks Practice Bank

100 Problems in 10 Categories

Neural Networks (CSE425 / EEE474 / CSE715)

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Instructions. Show all derivations for calculation problems. For conceptual items, answer in 2–4 sentences unless stated otherwise. Answers are not included in this handout.

1 Perceptron (Problems 1–10)

1. A perceptron has weights $\mathbf{w} = [2, -1]^\top$ and bias $b = -1$. For input $\mathbf{x} = [1, 3]^\top$, compute the pre-activation $z = \mathbf{w}^\top \mathbf{x} + b$ and the output $y = \text{sign}(z)$.

2. Explain why a single perceptron cannot represent the XOR function. Sketch the four XOR points in the (x_1, x_2) plane and argue geometrically.

3. Given training example ($\mathbf{x} = [0, 1]^\top$, $t = 1$) with current $\mathbf{w} = [1, 1]^\top$, $b = 0$, learning rate $\eta = 0.5$, perform **one** perceptron learning rule update. Use $y = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$.

4. True or False? "A perceptron with a linear activation $f(z) = z$ is equivalent to linear regression." Justify.

5. Write the perceptron convergence theorem assumptions. What guarantee does it provide when data are linearly separable?

6. For inputs in $\mathbb{R}^2$, describe the decision boundary $\mathbf{w}^\top \mathbf{x} + b = 0$ when $\mathbf{w} = [3, -4]^\top$, $b = 12$. Find the intercepts with both axes.

7. A dataset has labels $\{+1, -1\}$. After many epochs, the perceptron error never reaches zero. List **three** possible reasons.

8. Express the perceptron loss for misclassified point i using the hinge-style form $\max(0, -t_i(\mathbf{w}^\top \mathbf{x}_i + b))$ when $t_i \in \{-1, +1\}$.

9. Compare batch vs. online perceptron updates. Which tends to converge faster on noisy linearly separable data? Why?

10. Design a 2D two-class dataset with 6 points that **is** linearly separable. Write coordinates and a weight vector $\mathbf{w}$ that separates them.

2 Multilayer Perceptron (Problems 11–20)

11. A 2-layer MLP: input dim 2, hidden 3 with ReLU, output 1 with sigmoid. Count **total trainable parameters** (weights + biases).

12. For scalar output $\hat{y} = \sigma(z)$, $z = wx + b$, target $y \in \{0, 1\}$, derive $\partial \mathcal{L}/\partial w$ when $\mathcal{L} = -\big[y \log \hat{y} + (1 - y) \log(1 - \hat{y})\big]$.

13. Why is the XOR problem solvable with an MLP but not a single perceptron? Name the role of the hidden layer in one sentence.

14. Given ReLU activation, compute output for $\mathbf{z} = [-2,\ 0,\ 3]^\top$. Which units are "dead"?

15. Explain the **vanishing gradient** problem for deep MLPs with sigmoid activations. How does ReLU help?

16. An MLP uses tanh in the hidden layer and linear output for regression. Target $y = 5$, prediction $\hat{y} = 2$. With MSE $\mathcal{L} = \frac{1}{2}(y - \hat{y})^2$, what is $\partial \mathcal{L}/\partial \hat{y}$?

17. Draw a small MLP diagram for input $\mathbf{x} \in \mathbb{R}^4$, one hidden layer of size 5, output softmax over 3 classes. Label weight matrix shapes.

18. Universal approximation theorem (informal): what does it say about one hidden-layer MLPs? State one practical limitation.

19. You train an MLP and training loss decreases but validation loss increases after epoch 20. Diagnose and propose **two** fixes.

20. For multi-class classification with $C = 4$, write the softmax formula $\hat{y}_k$ and the cross-entropy loss for one-hot label $\mathbf{y}$.

3 Recurrent Neural Networks (Problems 21–30)

21. Write the vanilla RNN hidden-state update $\mathbf{h}_t = f(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b})$. Define each symbol.

22. Unroll an RNN for $T = 3$ time steps. Draw the computational graph showing shared weights across time.

23. Explain **exploding gradients** in RNNs. How does gradient clipping address it?

24. A character-level RNN predicts next letter. Sequence length $T = 50$, hidden size $H = 128$, vocab $V = 27$. Roughly how many parameters in $\mathbf{W}_{hh}$ and $\mathbf{W}_{hy}$ combined?

25. Why are vanilla RNNs poor at learning long-range dependencies? Relate to repeated multiplication of $\mathbf{W}_{hh}$ in BPTT.

26. Teacher forcing: define it and give one advantage and one disadvantage during training.

27. For sequence labeling, compare **many-to-one** vs. **many-to-many** RNN architectures. Give one application each.

28. BPTT with truncation at $\tau = 5$: describe what gradients are omitted and the trade-off.

29. If activation is tanh and all singular values of $\mathbf{W}_{hh}$ are 0.9, estimate $\|\partial \mathbf{h}_T / \partial \mathbf{h}_0\|$ order of magnitude for $T = 20$.

30. Write the RNN loss for a sequence as $\mathcal{L} = \sum_{t=1}^{T} \mathcal{L}_t$. How does backprop distribute $\partial \mathcal{L} / \partial \mathbf{W}_{hh}$ across time?

4 Long Short-Term Memory (Problems 31–40)

31. Write equations for LSTM gates: forget $\mathbf{f}_t$, input $\mathbf{i}_t$, candidate $\tilde{\mathbf{c}}_t$, cell $\mathbf{c}_t$, output $\mathbf{o}_t$, hidden $\mathbf{h}_t$.

32. If forget gate $\mathbf{f}_t \approx \mathbf{1}$ and input gate $\mathbf{i}_t \approx \mathbf{0}$, what happens to $\mathbf{c}_t$ vs. $\mathbf{c}_{t-1}$? Interpret.

33. Count trainable parameters for LSTM with input dim I, hidden H (standard formulation with four gates). Show your counting formula.

34. Explain how the **cell state** $\mathbf{c}_t$ enables gradient highway compared to vanilla RNN.

35. Peephole connections: what extra links do they add? Are they required in the standard LSTM?

36. Bidirectional LSTM: how are forward and backward outputs combined? Name one NLP task that benefits.

37. Stacking 2 LSTM layers: which output feeds the next layer at time t? Draw a 2-layer diagram.

38. LSTM for sentiment: input is word embeddings $T = 40$, hidden $H = 64$. What is the shape of $\mathbf{h}_{40}$ used for classification?

39. During BPTT, which LSTM gate gradients most often saturate sigmoid/tanh? What symptom during training?

40. Compare LSTM and vanilla RNN on copying a random token from position $t = 5$ to output at $t = 50$. Which wins and why?

5 Gated Recurrent Unit (Problems 41–50)

41. Write GRU update equations for reset gate $\mathbf{r}_t$, update gate $\mathbf{z}_t$, candidate $\tilde{\mathbf{h}}_t$, and hidden $\mathbf{h}_t$.

42. When $\mathbf{z}_t \approx \mathbf{0}$, how does $\mathbf{h}_t$ relate to $\mathbf{h}_{t-1}$? When $\mathbf{z}_t \approx \mathbf{1}$?

43. GRU vs. LSTM: which has fewer parameters for the same hidden size H? Give approximate parameter counts.

44. Role of **reset gate** $\mathbf{r}_t$ when computing $\tilde{\mathbf{h}}_t$: what past information is suppressed?

45. Map GRU gates intuitively to LSTM gates (which GRU mechanism resembles forget + input?).

46. Train GRU on sine wave prediction. Why might it outperform vanilla RNN with same H?

47. Implementational question: in PyTorch, `nn.GRU` returns `(output,` h_n`)`. What is the shape difference between them for batch-first input (B, T, I)?

48. Can a GRU suffer exploding gradients? Yes/no + one mitigation.

49. Design a tiny numeric example ($H = 1$) with $\mathbf{z}_t = 0.5$, $\mathbf{h}_{t-1} = 2$, $\tilde{\mathbf{h}}_t = 0$. Compute $\mathbf{h}_t$.

50. When would you choose GRU over LSTM in a production NLP pipeline? List **two** criteria.

6 Computational Graphs (Problems 51–60)

51. Define **operand node** and **operator node** in a computational graph. Give one example of each.

52. For $L = (y - \hat{y})^2$, $\hat{y} = wx$, draw the forward graph with nodes $\{w, x, \hat{y}, y, L\}$.

53. Using the graph in Problem 52, apply reverse-mode AD to find $\partial L/\partial w$ in terms of x, y, w.

54. Forward-mode vs. reverse-mode AD: which is preferred when the number of inputs $\ll$ number of outputs? Explain.

55. Chain rule on a graph: if $a \to b \to c \to L$, write $\partial L/\partial a$ as a product of local derivatives.

56. ReLU node $y = \max(0, x)$: what is $\partial y/\partial x$ for $x = -3$ and $x = 2$?

57. Sum node with inputs a, b, c and output $s = a + b + c$. What are $\partial s/\partial a$, $\partial s/\partial b$, $\partial s/\partial c$?

58. Explain **gradient accumulation** across a batch in terms of the graph over N samples.

59. A fork in the graph: x feeds both $u = x^2$ and $v = 3x$, then $L = u + v$. Compute $\partial L/\partial x$ at $x = 2$.

60. Why does PyTorch build a dynamic computational graph each forward pass? One advantage and one cost.

7 Batch Normalization (Problems 61–70)

61. For batch $\mathcal{B} = \{x_1, x_2, x_3, x_4\} = \{2, 4, 6, 8\}$, $\varepsilon = 0$, compute $\mu_{\mathcal{B}}$, $\sigma^2_{\mathcal{B}}$, and $\hat{x}_3$.

62. Write the BN affine transform $y_i = \gamma \hat{x}_i + \beta$. What happens when $\gamma = \sigma_{\mathcal{B}}$ and $\beta = \mu_{\mathcal{B}}$?

63. Derive $\partial L/\partial \beta$ given upstream gradients $\partial L/\partial y_i$.

64. Derive $\partial L/\partial \gamma$ in terms of $\hat{x}_i$ and $\partial L/\partial y_i$.

65. Why are γ and β learnable instead of fixing pure normalization?

66. Training vs. inference: what statistics are used at test time? Why not batch stats from a single test example?

67. Batch size $m = 1$: what problem arises for batch normalization? Name two fixes (conceptual).

68. Where is BN typically placed: before or after activation? Argue both sides briefly.

69. Internal covariate shift: define it and explain how BN mitigates it during training.

70. Mini-batch $\hat{x} = [-1, 0, 1, 2]$, $\gamma = 2$, $\beta = 1$, $\partial L/\partial y = [0.5, -0.5, 1, -1]$. Compute $\partial L/\partial \beta$ and $\partial L/\partial \gamma$.

8 Optimization (Problems 71–80)

71. SGD update with learning rate η: write θ_{t+1} in terms of $\nabla_\theta \mathcal{L}(\theta_t)$.

72. Momentum update with $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \nabla \mathcal{L}$, $\theta_{t+1} = \theta_t - \eta \mathbf{v}_{t+1}$. What role does β play?

73. Adam combines which two ideas? Write the first-moment and second-moment updates (symbols only).

74. Loss landscape sketch: identify saddle point, local minimum, and plateau. Why are plateaus slow for SGD?

75. Learning rate too high vs. too low: describe training loss behavior for each.

76. For $L(w) = (w-3)^2$, starting $w_0 = 0$, one step SGD with $\eta = 0.5$. Find w_1.

77. Explain **learning rate warmup** in transformer training. When is it useful?

78. Compare full-batch gradient descent vs. mini-batch SGD for large datasets. Memory and noise trade-offs.

79. Weight decay in AdamW: how does it differ from L2 regularization inside the loss for adaptive methods?

80. You observe loss oscillating wildly. List **three** hyperparameters to tune first and the direction of change.

9 Regularization (Problems 81–90)

81. L2 regularization: write the total loss $\mathcal{L}_{\text{total}}$ with penalty $\frac{\lambda}{2}\|\mathbf{w}\|_2^2$.

82. How does L2 penalty affect weight magnitudes during optimization? Geometric intuition in 2D.

83. L1 vs. L2: which promotes sparsity? Why?

84. Early stopping: describe the procedure using validation loss. What is the regularization effect?

85. Data augmentation as regularization: give **two** examples for images and **one** for text.

86. Bias-variance tradeoff: define both and show where regularization moves a model on this spectrum.

87. Label smoothing for softmax: replace hard target $y_k \in \{0, 1\}$ with what distribution? One benefit.

88. A model has train acc 99%, val acc 62%. Is this primarily high variance or high bias? What regularizers help?

89. Weight constraint / max-norm: cap $\|\mathbf{w}\|_2 \leq c$ after each step. How does this limit model capacity?

90. Bayesian view (informal): how does L2 prior on weights relate to Gaussian prior assumptions?

10 Dropout (Problems 91–100)

91. Dropout with rate $p = 0.5$: during training, what is the probability a hidden unit is **kept**?

92. Write the **inverted dropout** scaling: if mask $m_i \in \{0, 1\}$, how is output $\tilde{h}_i$ computed from h_i?

93. At inference, how are dropout masks handled? What scaling ensures train/test consistency?

94. Why does dropout reduce co-adaptation of hidden units? Intuitive explanation.

95. Dropout rate $p = 0.8$: is this light or heavy regularization? Expected fraction of dropped units?

96. Can dropout be applied to input layer? To output softmax layer? Discuss cautiously for each.

97. Monte Carlo interpretation of dropout at test time: what is the ensemble view?

98. Dropout vs. L2: both reduce overfitting. One scenario where dropout helps more.

99. Spatial dropout (concept): how does it differ from element-wise dropout in CNN feature maps?

100. Design an experiment: same MLP, with/without dropout $p = 0.3$. What three curves/tables will you report to judge improvement?

End of practice bank — 100 problems.